Problem 6: Detailed Solution

1. Elementary row operations

Apply elementary row operations to the matrix A, first eliminating the entries below the pivot in column 1:

Row operations: $R_2 \leftarrow R_2 - 3R_1, \quad R_4 \leftarrow R_4 - 5R_1$

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 3 & 2 & 1 & -3 & a \\ 0 & 1 & 2 & 6 & 3 \\ 5 & 4 & 3 & -1 & b \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & -1 & -2 & -6 & a-3 \\ 0 & 1 & 2 & 6 & 3 \\ 0 & -1 & -2 & -6 & b-5 \end{pmatrix}$$

Then add row 3 to row 2 and to row 4 respectively, eliminating the nonzero entries of those rows:

Row operations: $R_2 \leftarrow R_2 + R_3, \quad R_4 \leftarrow R_4 + R_3$

$$\longrightarrow \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & a \\ 0 & 1 & 2 & 6 & 3 \\ 0 & 0 & 0 & 0 & b-2 \end{pmatrix}$$

Swap rows 2 and 3, and arrange into standard row echelon form:

Row operation: $R_2 \leftrightarrow R_3$

$$\longrightarrow \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 6 & 3 \\ 0 & 0 & 0 & 0 & a \\ 0 & 0 & 0 & 0 & b-2 \end{pmatrix}$$

2. Determine the parameters a, b

The rank of the matrix is given as 2, and rows 1 and 2 are linearly independent, already forming the basis of rank 2.

For the matrix as a whole to have rank 2, rows 3 and 4 must be all-zero rows, so the constant terms at the end of these rows must be 0:

- Row 3: $a = 0$
- Row 4: $b - 2 = 0 \implies b = 2$

Final parameter values: $\boldsymbol{a = 0,\ b = 2}$

3. Find the maximal linearly independent subset of the set of column vectors

Substituting $a = 0,\ b = 2$, further reduce the row echelon form to reduced row echelon form:

Row operation: $R_1 \leftarrow R_1 - R_2$

$$\longrightarrow \begin{pmatrix} 1 & 0 & -1 & -5 & -2 \\ 0 & 1 & 2 & 6 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

The pivots (leading nonzero entries) lie in columns 1 and 2, so **columns 1 and 2** (denoted $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2$) form a maximal linearly independent subset of the set of column vectors.

4. Express the remaining vectors in terms of the maximal linearly independent subset

Reading the coefficients of the corresponding columns directly from the reduced row echelon matrix gives the linear expressions:

- Column 3: $\boldsymbol{\alpha}_3 = -\boldsymbol{\alpha}_1 + 2\boldsymbol{\alpha}_2$
- Column 4: $\boldsymbol{\alpha}_4 = -5\boldsymbol{\alpha}_1 + 6\boldsymbol{\alpha}_2$
- Column 5: $\boldsymbol{\alpha}_5 = -2\boldsymbol{\alpha}_1 + 3\boldsymbol{\alpha}_2$